

EX 16.2.2

STATES: Rain - Clean

$$P = \begin{matrix} & \begin{matrix} R & C \end{matrix} \\ \begin{matrix} R \\ C \end{matrix} & \begin{bmatrix} 0.5 & 0.5 \\ 0.1 & 0.9 \end{bmatrix} \end{matrix}$$

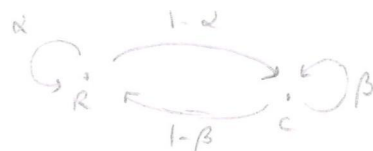
$$\frac{0.5}{2-0.5-0.9} = 0.1$$

$$\frac{1}{6} \quad \frac{1}{6}$$

EX 16.5.1

$$P = \begin{bmatrix} \alpha & 1-\alpha \\ 1-\beta & \beta \end{bmatrix}$$

$$\pi P = \pi$$



$$(\pi_R, \pi_C) \begin{bmatrix} \alpha & 1-\alpha \\ 1-\beta & \beta \end{bmatrix} = [\alpha \pi_R + (1-\beta) \pi_C, (1-\alpha) \pi_R + \beta \pi_C]$$

$$\begin{cases} \pi_R = \alpha \pi_R + (1-\beta) \pi_C \\ \pi_C = (1-\alpha) \pi_R + \beta \pi_C \\ \pi_R + \pi_C = 1 \end{cases}$$

$$(1-\alpha) \pi_R = (1-\beta) \pi_C$$

$$\pi_R = \frac{(1-\beta)}{(1-\alpha)} \pi_C$$

$$\frac{(1-\beta)}{(1-\alpha)} \pi_C + \pi_C = 1$$

$$(1-\beta) \pi_C + (1-\alpha) \pi_C = (1-\alpha)$$

$$\pi_C = \frac{(1-\alpha)}{(2-\beta-\alpha)}$$

$$\pi_R = \frac{(1-\beta)}{(2-\beta-\alpha)}$$

IN = OUT

$$R: \alpha \pi_R + (1-\beta) \pi_C = \alpha \pi_R + (1-\alpha) \pi_R$$

$$C: \beta \pi_C + (1-\alpha) \pi_R = \beta \pi_C + (1-\beta) \pi_C$$

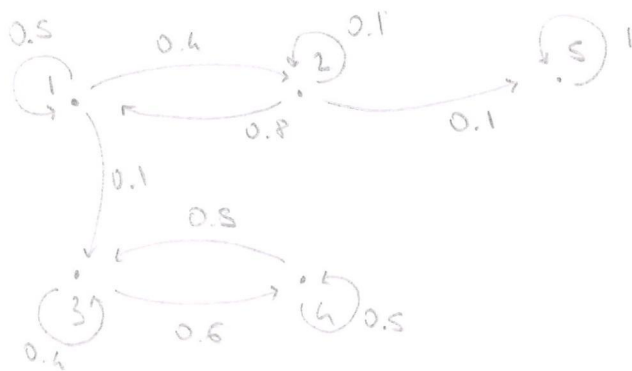
$$\pi_R + \pi_C = 1$$

$$\pi_R = \frac{(1-\beta)}{(1-\alpha)} \pi_C$$

(same as above)

$$\pi = \left(\frac{1-\alpha}{2-\beta-\alpha}, \frac{1-\beta}{2-\beta-\alpha} \right)$$

EXERCISE



$$P = \begin{bmatrix} 0.5 & 0.4 & 0.1 & 0 & 0 \\ 0.8 & 0.1 & 0 & 0 & 0.1 \\ 0 & 0 & 0.4 & 0.6 & 0 \\ 0 & 0 & 0.5 & 0.5 & 0 \\ 0 & 0 & 0 & 0 & 1 \end{bmatrix}$$

- WHAT IS THE PROBABILITY OF ENDING IN STATE 5 WHEN STARTING AT 1?
- FOR INITIAL STATE 1, CALCULATE THE LONG-RUN DISTRIBUTION
- HOW MANY TIMES WE WILL BE IN STATE 1 BEFORE WE LEAVE THE TRANSITION STATES?

• P_x prob of ending at S starting from x

$$\begin{cases} P_1 = 0.5 P_1 + 0.4 P_2 + 0.1 P_3 \\ P_2 = 0.8 P_1 + 0.1 P_2 + 0.1 P_5 \\ P_3 = 0 \\ P_4 = 0 \\ P_5 = 1 \end{cases}$$

$$P_2 = 0.8 P_1 + 0.1 P_2 + 0.1$$

$$P_2 = \frac{8}{9} P_1 + \frac{1}{9}$$

$$P_1 = \frac{1}{2} P_1 + \frac{4}{10} \left(\frac{8}{9} P_1 + \frac{1}{9} \right)$$

$$\frac{1}{2} P_1 = \frac{32}{90} P_1 + \frac{4}{90}$$

$$\frac{1}{2} - \frac{32}{90} = \frac{45-32}{90} = \frac{13}{90}$$

$$\frac{13}{90} P_1 = \frac{4}{90}$$

$$P_1 = \frac{4}{13} \quad \text{prob of ending at S starting from 1}$$

• $\pi_5 = \frac{4}{13}$ Done previous point

$$\Rightarrow \pi_3 + \pi_4 = 1 - \frac{4}{13}$$

$$\pi_3 + \pi_4 = \frac{9}{13}$$

$\pi_1 = \pi_2 = 0$ Absorbing states

$$P_{34} = \begin{bmatrix} 0.4 & 0.6 \\ 0.5 & 0.5 \end{bmatrix}$$

IN = OUT

$$0.6 \pi_3 = 0.5 \pi_4$$

$$\pi_3 = \frac{5}{11}$$

$$\pi_4 = \frac{6}{11}$$

$$\pi = \left(0, 0, \frac{5}{11}, \frac{6}{11}, \frac{4}{13} \right)$$

$$= \left(0, 0, \frac{45}{143}, \frac{54}{143}, \frac{4}{13} \right)$$

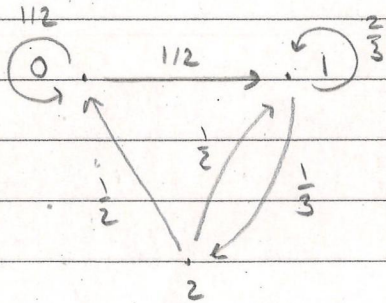
SANITY CHECK:

$$\frac{45}{143} + \frac{54}{143} + \frac{4}{13} \cdot \frac{11}{11} = \frac{45+54+44}{143}$$

$$= \frac{143}{143} = 1 \quad \checkmark$$

EXERCISE

Find steady-state probability



$$P = \begin{bmatrix} \frac{1}{2} & \frac{1}{2} & 0 \\ 0 & \frac{2}{3} & \frac{1}{3} \\ \frac{1}{2} & \frac{1}{2} & 0 \end{bmatrix}$$

$$\pi = \pi P$$

$$\pi P = [\pi_0 \quad \pi_1 \quad \pi_2] \begin{bmatrix} \frac{1}{2} & \frac{1}{2} & 0 \\ 0 & \frac{2}{3} & \frac{1}{3} \\ \frac{1}{2} & \frac{1}{2} & 0 \end{bmatrix}$$

1×3 3×3

$$= \left[\frac{1}{2}\pi_0 + \frac{1}{2}\pi_2, \frac{1}{2}\pi_0 + \frac{2}{3}\pi_1 + \frac{1}{2}\pi_2, \frac{1}{3}\pi_1 \right]$$

$$\begin{cases} \pi_0 = \frac{1}{2}\pi_0 + \frac{1}{2}\pi_2 \\ \pi_1 = \frac{1}{2}\pi_0 + \frac{2}{3}\pi_1 + \frac{1}{2}\pi_2 \\ \pi_2 = \frac{1}{3}\pi_1 \\ \pi_0 + \pi_1 + \pi_2 = 1 \end{cases}$$

$$\frac{1}{2}\pi_0 = \frac{1}{6}\pi_1, \quad \pi_0 = \frac{1}{3}\pi_1$$

$$\frac{1}{3}\pi_1 + \pi_1 + \frac{1}{3}\pi_1 = 1$$

$$\frac{4}{3}\pi_1 = 1 \quad \pi_1 = \frac{3}{4}$$

$$\pi_0 = \frac{1}{3}$$

$$\pi_2 = \frac{1}{4}$$

$$\pi = \left(\frac{1}{3}, \frac{3}{4}, \frac{1}{4} \right)$$

EXERCISES on TIME

IMAGINE WE DO NOT SPEND THE SAME AMOUNT OF TIME IN EACH OF THE STATES. CALCULATE THE STEADY STATE DISTRIBUTION.

1.



BALANCE CD
CHAPTER 16.8
OF HILLIER-UEBERMAN

WITHOUT TIME DIFFERENCE:

IN = OUT

$$\frac{1}{2}\pi_1 + \frac{1}{2}\pi_2 = \frac{1}{2}\pi_1 + \frac{1}{2}\pi_1 \quad \text{STATE 1}$$

$$\Rightarrow \pi_1 = \pi_2$$

$$\pi_1 + \pi_2 = 1$$

$$\Rightarrow \pi = \left(\frac{1}{2}, \frac{1}{2}\right)$$

WITH TIME DIFFERENCE:

$$\left(\frac{1}{2} \cdot 1, \frac{1}{2} \cdot 2\right) = \left(\frac{1}{2}, 1\right)$$

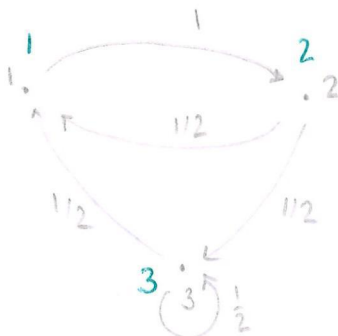
$$= \left(\frac{1}{2}, \frac{2}{2}\right)$$

LOOKING AT THE
NUMERATOR WE
HAVE 3 TIME
UNITS TO "ALLOCATE"

$$\text{NORMALIZING: } \pi_T = \left(\frac{1}{1+2}, \frac{2}{1+2}\right) = \left(\frac{1}{3}, \frac{2}{3}\right)$$

WE SPEND $\frac{1}{3}$ OF THE TOTAL TIME IN 1
and $\frac{2}{3}$ IN 2

2.



WITHOUT TIME DIFFERENCE:

IN = OUT

STATE 1

$$\frac{1}{2}\pi_2 + \frac{1}{2}\pi_3 = \pi_1$$

STATE 2

$$\pi_1 = \frac{1}{2}\pi_2 + \frac{1}{2}\pi_2$$

$$\pi = \left(\frac{1}{3}, \frac{1}{3}, \frac{1}{3}\right)$$

STATE 3

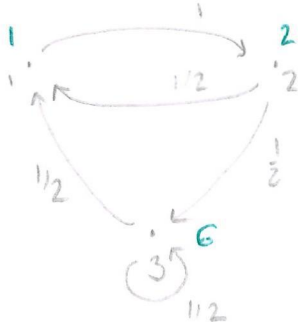
$$\frac{1}{2}\pi_3 + \frac{1}{2}\pi_2 = \frac{1}{2}\pi_3 + \frac{1}{2}\pi_3$$

$$\pi_1 + \pi_2 + \pi_3 = 1$$

$$\text{WITH TIME DIFFERENCE: } \left(\frac{1}{3} \cdot 1, \frac{1}{3} \cdot 2, \frac{1}{3} \cdot 3\right) = \left(\frac{1}{3}, \frac{2}{3}, \frac{3}{3}\right)$$

$$\text{NORMALIZING: } \left(\frac{1}{1+2+3}, \frac{2}{1+2+3}, \frac{3}{1+2+3}\right) = \left(\frac{1}{6}, \frac{1}{3}, \frac{1}{2}\right)$$

3.



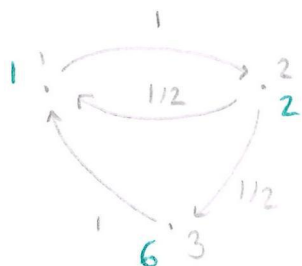
WITHOUT TIME DIFFERENCE: $\pi = \left(\frac{1}{3}, \frac{1}{3}, \frac{1}{3}\right)$ (SAME AS BEFORE)

WITH TIME DIFFERENCE: $\left(\frac{1}{3} \cdot 1, \frac{1}{3} \cdot 2, \frac{1}{3} \cdot 6\right) = \left(\frac{1}{3}, \frac{2}{3}, \frac{6}{3}\right)$

$1+2+6 = 9$ TIME UNITS

$$\pi_T = \left(\frac{1}{9}, \frac{2}{9}, \frac{6}{9}\right)$$

4.



WITHOUT TIME DIFFERENCE:

IN = OUT

STATE 1

$$\frac{1}{2} \pi_2 + \pi_3 = \pi_1$$

STATE 2

$$\pi_1 = \frac{1}{2} \pi_2 + \frac{1}{2} \pi_2$$

$$\pi_1 = \pi_2$$

STATE 3

$$\frac{1}{2} \pi_2 = \pi_3$$

$$\pi_2 = 2\pi_3$$

$$\pi_1 + \pi_2 + \pi_3 = 0$$

$$2\pi_3 + 2\pi_3 + \pi_3 = 0$$

$$\pi_3 = \frac{1}{5}$$

$$\pi = \left(\frac{2}{5}, \frac{2}{5}, \frac{1}{5}\right)$$

WITH TIME DIFFERENCE: $\left(\frac{2}{5} \cdot 1, \frac{2}{5} \cdot 2, \frac{1}{5} \cdot 6\right) = \left(\frac{2}{5}, \frac{4}{5}, \frac{6}{5}\right)$

$2+4+6 = 12$ TIME UNITS

$$\pi_T = \left(\frac{2}{12}, \frac{4}{12}, \frac{6}{12}\right) = \left(\frac{1}{6}, \frac{1}{3}, \frac{1}{2}\right)$$